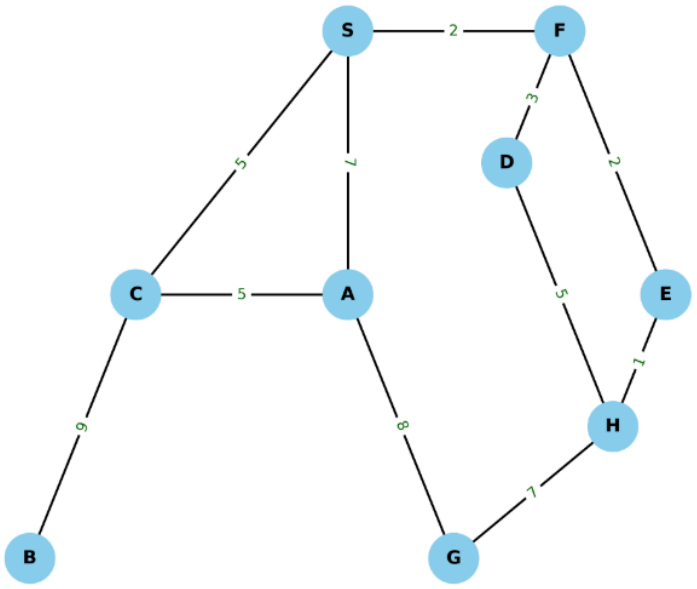
CSE422: Artificial intelligence

Assignment 01 [Summer 2026] (v1.0)

Informed Search

1. For the following graph, **S** is the **starting node**, and **G** is the **goal node.** Heuristic values are given in the table on the right.



Node	Heuristic	Node	Heuristic
S	12	D	8
A	14	E	15
B	16	F	9
C	10	H	5

- Simulate Greedy Best-first search (GBFS) on the graph to find out **the path to the goal**, as well as **the cost of the path.**
- Simulate A-star search to find the optimal path to the goal, and compare it to GBFS output.

2. Suppose you have three heuristic functions h1, h2 and h3. Among these h1 and h2 are admissible but h3 is inadmissible. You have decided to create several new heuristic functions defined as follows:

- h4(n) = 0
- h5(n) = 2 × h2(n)
- h6(n) = (h1(n)+h2(n))/2
- h7(n) = max(h1(n), h2(n))
- h8(n) = min(h1(n), h2(n))
- h9(n) = min(h1(n), h3(n))
- h10(n) = max(h2(n), h3(n))

Now answer the following questions and justify.

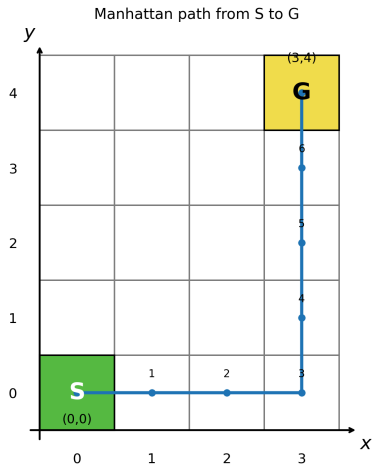
a. For each heuristic, determine whether it is admissible or not.
b. Among h7 and h8 which one is dominant (better)?
c. In your opinion which heuristic is the best?

3. Imagine you have a **6×6** grid, where an agent's goal is to move from the **top-left corner** (0, 0) to the **bottom-right corner** (5, 5). The movement costs are: **horizontal or vertical = 1, diagonal = √2.** You are given TWO heuristics:

- $h_1(n)$ = **Manhattan distance** from node **n** to goal
- $h_2(n)$ = **Euclidean distance** from node **n** to goal

Which of these heuristics will expand fewer nodes in this grid? Explain.

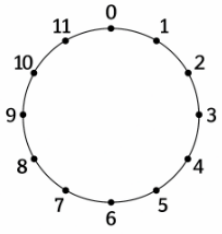
4.



Consider the given 4×5 grid. Movement is allowed in horizontal and vertical directions only. Diagonal movement is not allowed. The cost of each move is 1. Let the start node be S=(0,0) and the goal node be G=(3,4). We consider the Manhattan distance as a heuristic for this problem: $h_M(n) = |x_n - x_G| + |y_n - y_G|$

a. Find the shortest path from S to G and its cost (no need to use any algorithm, just use observation).
b. Suggest another heuristic for this problem, $h_E(S)$.
c. Suppose N is the node with coordinates (1, 2). Show if the heuristics $h_E(S)$ and $h_M(S)$ are admissible forTHIS node. If a heuristic is admissible for one node, can we say that heuristic is admissible in general for the problem?
d. Determine whether $h_E(S)$ and $h_M(S)$ are consistent for the STARTING node S.

5.



Consider a circular grid with **12** nodes labeled **[0,1,2,.....,11]** arranged like the numbers on a clock. From any node (n), the agent can move:

- **Clockwise to node ((n+1) mod 12) with cost 1**
- **Anticlockwise to node ((n-1) mod 12) with cost 2**

Starting node: [5]; goal node: [0]

A heuristic function **(h(n))** is proposed for use with **A* graph search**, defined as: **h(n) = min(2n, (12 - n))**

a. List the first THREE nodes that will be visited/expanded/popped using A* search on this grid. Ties are broken in favor of the **clockwise** successor.
b. Determine whether h(n) is admissible in this graph by examining any TWO nodes.
c. Determine whether h(n) is consistent in this graph by examining any TWO sets of nodes.

Local Search (Hill Climbing + Simulated Anealing)

6. Suppose you are inside a grid-based maze, where each grid holds a treasure. The value of the treasure is given by an integer. When you are on a grid, you can collect the treasure of that grid, and you can also inspect the treasures in all neighbouring grids (i.e., the grids with which your current grid shares an **edge** or a **corner**). If you start from any of the **corner grids** (i.e. top-left, bottom-left, top-right, or bottom-right), can you find the highest valued treasure hidden in the maze using the **hill-climbing algorithm**? Explain why or why not. You can use the following grid given as an example.

1	4	3	2
6	10	5	1
2	9	8	4
3	1	8	15
5	7	3	6
4	6	5	10

7.

$$f(x) = \begin{cases} 100 - (x - 0.3)^2, & \text{if } (0.2 \leq x < 0.6) \text{ or } (0.7 < x \leq 0.9), \\ 99.91 + 0.1x, & \text{if } 0.6 \leq x \leq 0.7. \end{cases}$$

Suppose Frank is given the above piecewise function **f(x)** where **x ∈ [0.2, 0.9]**. He needs to find the value of **x** that **maximizes** the function **f(x)**. Frank decided to apply the Hill Climbing algorithm for this task, using the values **{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9}** for x**.**

a. Sketch the function with the tried values and explain whether Frank will encounter a **local maxima**.
b. Explain a local search technique that Frank could have used to alleviate the problem in (a).

8. Now consider **minimizing** the function **f(x) = min(95, − (x−0.3)^2 + 100)** where **x** can be any integer from **-1 to 5**. Explain whether Hill Climbing will encounter **a plateau** during the search.

9. Explain how the **temperature** parameter controls the probability of neighbour selection in simulated annealing.

10. State whether the following statements are TRUE or FALSE with an explanation. A short explanation is compulsory, just writing TRUE or FALSE will not yield any mark.
a. Simulated annealing's probability of denying worse solutions increases as the search progresses.
b. Simulated annealing guarantees better performance than hill climbing in all types of search spaces.
c. Simulated annealing becomes equivalent to hill climbing when the initial temperature is very high.
d. When optimizing a function with many local maxima of similar values, random restarts combined with hill climbing will never increase the likelihood of finding the global optimum.
e. A hill-climbing algorithm that stops after a fixed number of iterations will never find the global optimum if the number is large enough.

11. Problem Statement: Select food items to pack in the lunchbox so that the total volume does not exceed 1liter and the total nutritional value is maximized.
Lunchbox Volume Limit: **1 liters**
Food Items:

- Sandwich: Volume = 0.5 liters, Nutritional Value = 60
- Apple: Volume = 0.3 liters, Nutritional Value = 30
- Yogurt: Volume = 0.5 liters, Nutritional Value = 50
- Granola Bar: Volume = 0.2 liters, Nutritional Value = 40
- Carrot Sticks: Volume = 0.3 liters, Nutritional Value = 35

a. Encode the problem, so it can be solved using any **local search algorithm such as Hill Climbing.**
[Hint: Encoding is similar to 0/1 knapsack genetic algorithm encoding, i.e. a string of binaries. However, in local search we need to have a START state. What will be the start state for this problem? When we choose no food item at all.]
b. Explain how you would generate neighboring solutions for this local search method.

[Hint: For each state, a neighboring state can be found by adding one more food item. However, if we are at maximum volume limit, we can not find any neighboring states.]

c. Using the given problem setup, demonstrate the main difference between Simulated Annealing and Hill Climb algorithm? Assume, Temperature parameter = 100 currently

Genetic Algorithm

12. You are a delivery driver who must visit **8 cities exactly once** and return to the starting point. Each pair of cities has a known **distance** between them, as given in the following table. Your goal is to find the **shortest possible route** that visits all cities once and returns to the start. Use **Genetic Algorithm** to solve this problem.

	A	B	C	D	E	F	G	H
A	-	4	8	6	7	3	5	9
B	4	-	2	7	3	6	8	5
C	8	2	-	9	4	7	6	3
D	6	7	9	-	2	4	5	7
E	7	3	4	2	-	6	3	5
F	3	6	7	4	6	-	2	8
G	5	8	6	5	3	2	-	4
H	9	5	3	7	5	8	4	-

- Propose an encoding scheme to represent candidate solutions of the problem.
- Generate an initial population of 6 using the proposed encoding scheme.
- Propose a fitness function suitable for the problem.
- Run one iteration of Genetic Algorithm for this problem. Choose random crossover and mutation indices.
- Show the best solution after running one iteration.
- Show what the population for the next iteration will be using 2-elitism.

13. You are given a 4th order polynomial function: $f(x) = x^4 - 14x^2 + 24x + 30$. This function has many local maxima and minima, but our target is to find the global minimum value of the function. We will use Genetic algorithm for this.

- Propose an encoding scheme to represent candidate solutions of the problem. [Hint: your encoding must be such that it is possible to perform operations like cross-over and mutation. You could say x1 = 5, x2 = 6, … is a valid encoding scheme, but how would we perform cross-over or mutation? So, as you can see, it is not actually a good encoding.]
- Generate an initial population of 4 using the proposed encoding scheme.
- Propose a fitness function suitable for the problem.
- Run one iteration of Genetic Algorithm for this problem. Choose random crossover and mutation indices.
- Show the best solution after running one iteration.
- Show what the population for the next iteration will be using 2-elitism.

14. You are tasked with designing a seating arrangement for a wedding reception with 8 guests. The guests have specific preferences about whom they would like to sit next to and whom they want to avoid. You need to create a seating arrangement for these guests around a circular table. Depending on the guests' preferred seating arrangements, the aim is to optimize their level of satisfaction. Consider the following constraints:

- Guest 1 (G1) prefers to sit next to Guest 2 and Guest 3 but wants to avoid Guest 4.
- Guest 2 (G2) prefers to sit next to Guest 1 and Guest 4 but wants to avoid Guest 5.
- Guest 3 (G3) prefers to sit next to Guest 1 and Guest 6 but wants to avoid Guest 7.
- Guest 4 (G4) prefers to sit next to Guest 2 and Guest 5 but wants to avoid Guest 8.
- Guest 5 (G5) prefers to sit next to Guest 4 and Guest 6 but wants to avoid Guest 1.
- Guest 6 (G6) prefers to sit next to Guest 3 and Guest 5 but wants to avoid Guest 2.
- Guest 7 (G7) prefers to sit next to Guest 8 but wants to avoid Guest 3.
- Guest 8 (G8) prefers to sit next to Guest 7 but wants to avoid Guest 4.

You have to solve the problem using genetic algorithm.

- Encode the problem and create four parent chromosomes.
- Suggest a possible fitness function and choose the two fittest Chromosomes.
- Do you think the regular crossover (as in the n-queen problem) will work in this scenario? Explain why. [Hint: this is a permutation problem, just like the TSP one we saw in class. Each guest should only appear once in a permutation, but this constraint may be violated if we use regular crossover. We saw how to use an appropriate crossover for these cases in class.]
- Show how to perform mutation in this context.

15. You are the surveillance officer of a museum. You want to bring the museum under CC-TV surveillance. You have THREE cameras, but you need to cover 20 distinct areas. Each camera has a CAPACITY, i.e. how many places (both towards its left and right) can a camera cover. For example, a camera with a CAPACITY of 3 can cover 3 places to its left, 3 places to its right, and the spot where it is placed (so altogether 7 places). The first two cameras have a CAPACITY of 2, and the remaining camera has a CAPACITY of 3. Based on this, you need to place the cameras at strategic positions so you can cover as many areas as possible. You will use genetic algorithm for this.

a. Design an encoding scheme for the problem.
b. Design a suitable fitness function.
c. Using your encoding scheme, show FOUR parents and calculate their fitness.
d. Will single point crossover work for this problem? Explain. [Hint: NO]

16. A student wants to plan their daily routine (8 AM – 10 PM) by arranging **five** activities in an optimal sequence. Each activity can be performed at most once per day. Available activities: **Study**, **Eat**, **Nap**, **Exercise**, **Entertainment**, **Socializing.** The goal is to maximize productivity, health, and happiness.

Activity Constraints

A. Avoid (soft constraints)

- **Studying** immediately after **Eating** can cause laziness
- **Exercising** immediately after **Eating** can be uncomfortable
- **Entertainment** before **Study** early in the day can cause distraction
- **Socializing** right after waking up can waste time

B. Must NOT violate (hard constraints)

- Must include **Study** and **Eating** activity
- **Exercise** cannot be the last activity of the day

C. Activity Priority

- **High**-priority: **Study, Eat**
- **Medium**-priority: **Exercise**
- **Low**-priority: **Entertainment**, **Nap**, **Socializing**

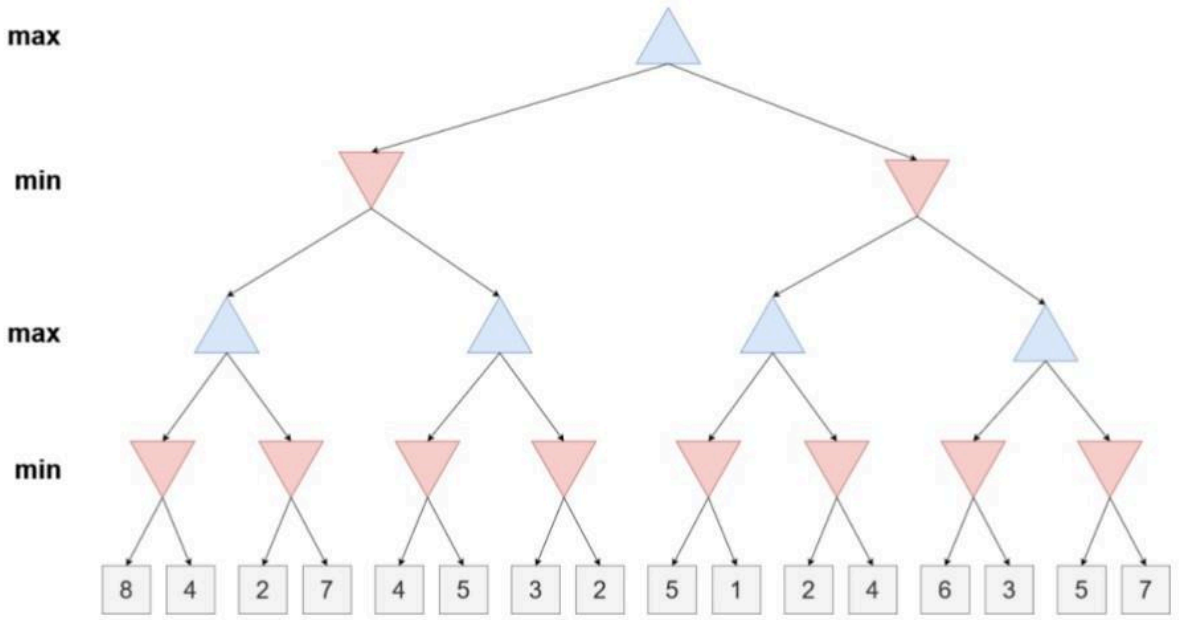
Note: Performing high-priority activities earlier improves fitness.

a. **Explain the encoding scheme** for the problem and create four parent chromosomes.
b. **Suggest** a suitable fitness function (can be explained in words as well, instead of writing the function), and **check** the fitness of any chromosome you showed in part **a**.
c. Will single-point crossover work in this case? Why or why not?

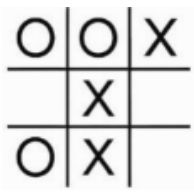
Adversarial Search (Alpha-beta Pruning)

17. Apply the Minimax algorithm with **alpha-beta** pruning on the following game tree to determine the best possible outcome for the MAX player at root node, assuming both MAX and MIN play optimally.

- Show alpha and beta values for each node.
- Show which branches can be pruned.

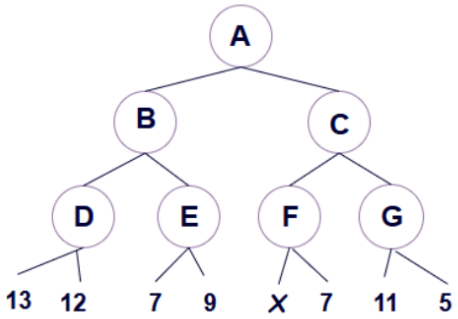


18. Suppose you are to design an **agent (circle)** that knows how to play a game of tic tac toe against the **user (cross)**. In THIS problem, the circle is the first move (agent goes first). The following figure shows the current state of the game. Now the agent has to decide which move to make by using the minimax algorithm with alpha beta pruning.



Determine which move would the agent prefer from the given state of the game. The final utilities will be [(100 - Depth) * 1] for a win, [(100 - Depth) * -1] for a loss and 0 for a draw. You must showcase the values of the nodes, α and β.
[Hint: Assuming the depth of the root node is 1, if a leaf node is found at the next level (depth 2) where the agent wins, then the utility for that leaf node from the agent's perspective would be (100 - 2) *1 = 98. For a losing state it would have been (100 - 2) * -1 = -98]

19. Simulate the following state space tree graph using Alpha Beta pruning and generate a value of 'X', such that **the branch from state C to G does not get pruned.**



20. A row of coins is placed on the table with values in a specified order: **[8, 15, 3, 7].**
Two players (an AI agent and a human) take turns picking coins under the following rules:

- On each turn, a player can pick only one coin, either the leftmost or the rightmost of the row. For example, on the first turn from [8, 15, 3, 7], the player can choose either 8 (leftmost coin) or 7 (rightmost coin). And this rule will be followed throughout the game. The order of the coins will remain unchanged.
- The game continues until all coins are picked.
- Each player tries to maximize the total value of coins they collect and, in turn, minimize the total value of coins the opponent collects.
- The agent will make the first move.
- The utility function, U(s) = (Total coin value of myself − Total coin value of enemy), along a certain path.

Suppose, YOU are the AI agent. Your goal is to choose a sequence of optimal moves.

a. Draw the full game tree, following the rules stated above.
b. Choose the optimal move for the agent using the Minimax algorithm with Alpha-Beta pruning. You must showcase the values of alpha and beta, as well as the pruned branches (if there are any).

FORTUNE FAVORS THE BRAVE